CS1910 Computer Science I • Computational Thinking **Partner workbook**

Describe and Draw

In lecture with a partner • about 20 minutes

Name: _____ **Partner:** _____

This PDF can be filled in: click in an answer space to type, or print it and write. Draw diagrams with your PDF viewer's pen or markup tool, or on paper.

What this workbook practises

Before we write any code, we need to be able to describe a problem precisely. In this activity one partner describes a picture, using words only, and the other draws it. Then you will use the ideas of computational thinking (decomposition, pattern recognition, abstraction and algorithms) to explain what made the description work.

1 Describe and draw

about 5 minutes

How it works

- One partner is the **communicator**; the other is the **illustrator**.
- The communicator looks at the picture on the **last page**, and does **not** show it to the illustrator.
- The illustrator draws in the box below, and does **not** show the drawing to the communicator.
- Using **only spoken instructions**, the communicator tells the illustrator what to draw. Only the communicator may talk.
- You have **5 minutes**. Then compare the drawing with the picture.

Illustrator: draw here

Illustrator's drawing

2 Debrief

about 10 minutes

1. Which instructions worked well? Which were confusing? Give one example of each.

2. **Decomposition:** into which separate parts would you break this picture?

3. **Pattern recognition:** what repeats in the picture? How could you describe the repeated part once?

4. **Abstraction:** which details really matter to recreate the picture, and which can be ignored?

5. **The illusion:** what are the key features of this picture that make the red lines look curved?

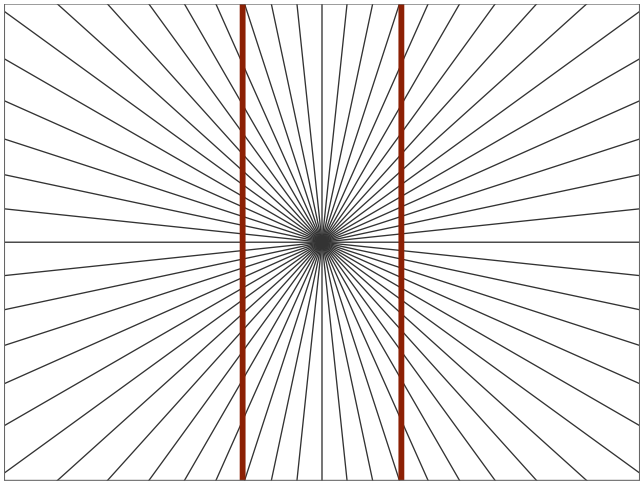
6. **Algorithm:** write the steps a computer could follow to draw this picture.

Before you leave

about 2 minutes

- [ ] I can break a picture into parts (decomposition) and spot what repeats (pattern recognition).
- [ ] I can decide which details matter and which can be ignored (abstraction).
- [ ] I can write steps precise enough for someone else, or a computer, to follow (algorithm).

Communicator only: do not show this page to your partner



Describe this picture to your partner using words only. Do not point, gesture or show them the page.